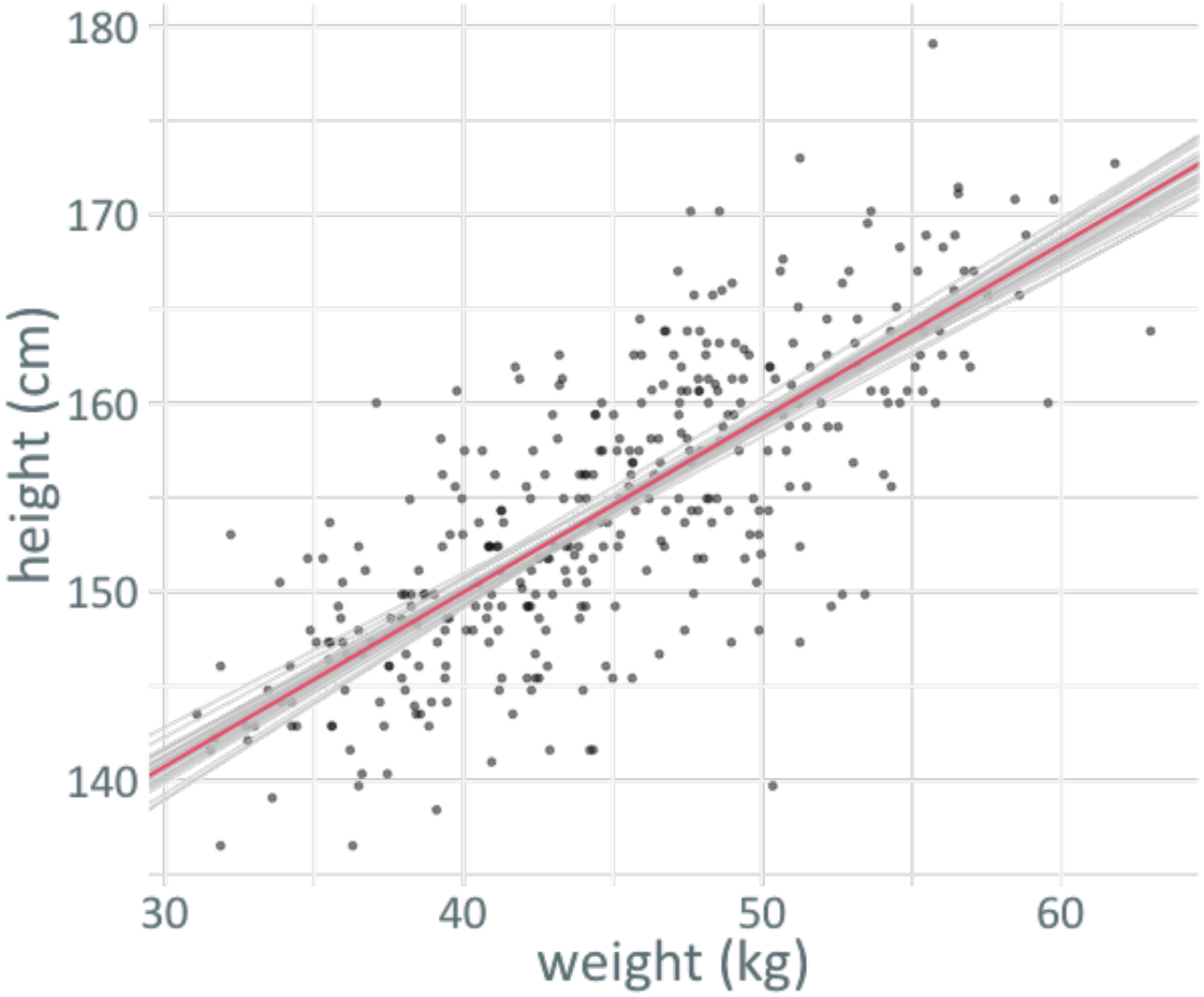
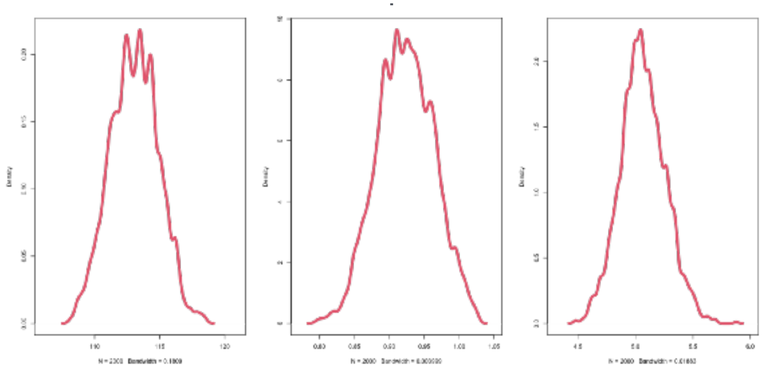
Modelo altura-peso

$$y_i \sim Normal(\mu_i, \sigma)$$
$$\mu_i = \alpha + \beta x_i$$
$$\alpha \sim Normal(0, 20)$$
$$\beta \sim lognormal(0, 1)$$
$$\sigma \sim Exponential(1)$$



Amostradores no Rethinking

`quap()` – aproximação quadrática



$$y_i \sim Normal(\mu_i, \sigma)$$
$$\mu_i = \alpha + \beta x_i$$
$$\alpha \sim Normal(0, 20)$$
$$\beta \sim lognormal(0, 1)$$
$$\sigma \sim Exponential(1)$$

```
# Data
library(rethinking)
d2 <- Howell1[ Howell1$age >= 18 , ]

# Model
quap(alist(
  y ~ dnorm(mu, sigma),
  mu <- a + b * x,
  a ~ dnorm(0, 20),
  b ~ dlnorm(0, 1),
  sigma ~ dexp(1)),
  data = list(y = d2$height,
              x = d2$weight))
```